DS7010 – Data Science Dissertation

Title: Explainable Spatial Machine Learning For House Price Prediction In London Using Open Data

by

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Abstract

House price prediction is an important problem in both research and practice because property values influence the household decisions, investment activity, mortgage risk and urban planning. In a city such as London, house price prediction is very challenging because prices are shaped not only by property characteristics, but also by spatial and neighbourhood conditions such as accessibility, centrality and deprivation. This dissertation investigates whether machine learning models combined with open spatial and socio-economic data can be used to predict residential house prices in London more effectively.

The study uses HM Land Registry Price Paid Data for 2022, 2023 and 2024, enriched with postcode geography from the ONS Postcode Directory, neighbourhood deprivation measured through IMD decile, and transport accessibility derived from Transport for London stop and station data. Additional engineered features include distance to the nearest station and distance to Central London. After data cleaning, staged construction and filtering, the final modelling dataset contained 388,881 observations. A time-based split was used, with 2022–2023 as the training set and 2024 as the test set, to reflect a more realistic forecasting setting.

Three regression models were compared: ElasticNet, Random Forest and Histogram Gradient Boosting. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2, Median Absolute Error and the percentage of predictions within 10% of the true sale price. The results showed that Random Forest was the strongest overall model. After light hyperparameter tuning, it achieved a MAE of £191,716.64, RMSE of £502,547.67, R^2 of 0.4856, Median Absolute Error of £80,728.21, and 33.09% of predictions within 10% of the true price on the 2024 test set.

The findings also showed that spatial and neighbourhood-based feature engineering improved the modelling framework. In particular, distance to Central London emerged as the most important predictor in the final model, followed by tenure type, geographic coordinates, property type and IMD decile. However, the analysis also showed that prediction accuracy was weaker for high-value and unusual properties, reflecting both the heterogeneity of luxury housing and the absence of richer property-level variables such as floor area, room count and condition.

Overall, the dissertation demonstrates that open data spatial machine learning provides a useful and interpretable framework for London house price prediction. It also shows that the main gains came from thoughtful dataset construction and feature engineering rather than extensive hyperparameter optimisation alone.

1. Introduction

1.1 Background

House price prediction is an important topic in both academic research and practical decision making. For households, purchasing a home is often one of the largest financial commitments they will make, while for investors, lenders and policymakers, changes in property values can influence investment decisions, mortgage risk, housing affordability and urban planning. However, predicting house prices is difficult because property values are shaped by a combination of structural, locational and socio-economic factors, and these influences often interact in complex ways. In a city such as London, this complexity is even greater because of strong variation in neighbourhood quality, transport accessibility, deprivation, demand and overall market conditions.

Traditional property valuation has often drawn on hedonic pricing ideas, where a house is valued according to the combined effect of its different characteristics. This provides an important theoretical foundation, but real housing markets are often too complex to be explained fully by simple linear relationships. In practice, the relationship between housing characteristics and price is frequently non-linear, and location can have a particularly strong effect. Two properties with similar physical characteristics may still have very different prices if one is located in a more central, better connected or less deprived area. This makes machine learning a useful approach because it can capture more complex patterns than conventional regression models.

This dissertation focuses on predicting residential house prices in London using open data and machine learning. The study combines official property transaction data with postcode geography, neighbourhood deprivation and transport accessibility variables in order to build a more spatially informed prediction model. Rather than relying only on basic transaction information, the project aims to show how location-based and neighbourhood-based factors can improve the modelling of house prices.

1.2 Problem Statement

Although many studies have examined house price prediction, important challenges remain. One of the main issues is that property prices are not determined only by the dwelling itself, but also by the wider context in which it is located. This includes transport connectivity, centrality, neighbourhood deprivation and other local influences. If such factors are ignored, prediction models may fail to capture important parts of the housing market.

Another challenge is that some predictive studies focus strongly on accuracy while giving less attention to realistic evaluation and model interpretation. In practice, a useful model should not only perform well, but should also be tested in a setting that reflects future prediction and should provide some understanding of why it performs as it does. For this reason, the present dissertation uses a time-based train-test split and includes model interpretation through visual analysis and feature importance.

1.3 Aim and Objectives

The main aim of this dissertation is to develop and compare machine learning models for predicting house prices in London using open data, with a particular focus on spatial and neighbourhood-related features.

The objectives of the study are:

- to build a London house price dataset using official open data sources;
- to enrich the transaction data with spatial and socio-economic variables;
- to compare the performance of multiple machine learning regression models;
- to evaluate the models using appropriate regression metrics and a realistic time-based split;
- to interpret the behaviour of the best-performing model using visualisations and feature importance analysis.

1.4 Research Questions

This dissertation is guided by the following research questions:

1. Which machine learning model performs best for predicting London house prices using the selected open datasets?
2. Do spatial and socio-economic features improve predictive performance?
3. Which variables appear to have the strongest influence on the predictions produced by the best-performing model?

1.5 Structure of the Dissertation

The remainder of the dissertation is organised as follows. Chapter 2 reviews the literature on hedonic pricing, spatial influences on housing value, transport accessibility, machine learning approaches to house price prediction and model explainability. Chapter 3 explains the methodology, including the datasets used, data preparation, feature engineering, modelling approach and evaluation metrics. Chapter 4 presents the results of the model comparison and the main visual findings. Chapter 5 discusses the results in relation to the research questions and the wider literature, and Chapter 6 concludes the dissertation by summarising the main findings, limitations and possible directions for future work.

1. Literature Review

2.1 Introduction

House price prediction has attracted considerable attention in both academic research and professional practice because housing is one of the most important asset classes in modern economies. Property values affect households, investors, lenders and policymakers, and accurate estimation is useful for valuation, lending decisions, taxation and urban planning. At the same time, house prices are difficult to model because they are influenced by many interacting factors, including structural property characteristics, local accessibility, neighbourhood quality and broader market conditions. In a city such as London, these influences are especially complex because price levels vary sharply across areas and because location can affect value in multiple ways at once. For this reason, the literature on house prices has developed across several connected areas, including hedonic pricing theory, spatial modelling, transport accessibility, machine learning and model explainability.

2.2 Hedonic pricing as the theoretical foundation

A major starting point in housing research is hedonic pricing theory. Rosen (1974) argued that the price of a differentiated good can be understood as the combined value of its individual characteristics. In the housing context, this means that a house is not priced only as a single object, but as a bundle of attributes such as size, type, tenure, accessibility, environmental quality and location. This idea remains important because it provides the conceptual basis for feature-based house price models. Even when machine learning is used instead of classical regression, the underlying assumption remains that price can be predicted from observable property and neighbourhood characteristics. In this sense, modern predictive modelling continues to build on the hedonic tradition rather than replacing it.

However, traditional hedonic approaches were often estimated using relatively simple linear specifications. In real housing markets, price relationships are rarely so straightforward. Some variables may have non-linear effects, while others may interact in ways that are difficult to capture using a basic regression equation. Can (1992) highlighted this issue by comparing traditional and spatial autoregressive hedonic models and showing that modelling choices matter for how housing price determination is understood.

This is important for the present dissertation because it supports the decision to move beyond a purely linear framework when modelling London house prices.

2.3 Spatial dependence and locational effects

One of the clearest findings in the literature is that housing markets are spatial. Property values are affected not only by a dwelling's own features, but also by the characteristics of nearby places. Houses located close to one another often share access to the same schools, transport links, environmental quality and local reputation, meaning that their prices are not fully independent. Can and Megbolugbe (1997) showed that spatial dependence in housing datasets can affect the precision and accuracy of price estimation and index construction. This is particularly relevant in urban markets, where strong local clustering is common. Their work suggests that ignoring spatial structure can weaken the reliability of house price models.

This idea is reinforced by the broader field of spatial econometrics. LeSage (2008) explains that spatial econometric methods were developed to account for dependence between nearby observations and to extend conventional regression models in a more realistic way. In property research, this is useful because price patterns often follow geographic structure rather than appearing randomly across space. Orford (2000) and Orford (2002) also demonstrated the importance of spatial structure in local housing markets, arguing that location-sensitive and multilevel approaches can improve the modelling of local price dynamics and locational externalities. Together, these studies support the view that location should be treated as a central component of housing value rather than as a background variable.

For the present dissertation, this literature is especially important because the study aims to predict London house prices using spatially enriched features. The inclusion of latitude, longitude, distance to Central London and distance to the nearest station is therefore not simply a technical decision, but one supported by the wider theory that housing values are closely linked to geographic context.

2.4 Accessibility, transport and neighbourhood context

Transport accessibility is another major theme in the house price literature. A long line of studies has shown that rail or transit investment can influence residential values. Dewees (1976) examined the effect of a subway on residential property values in Toronto and provided early evidence that transit infrastructure can affect housing markets. Bowes and Ihlanfeldt (2001) later identified the impacts of rail transit stations on residential property values, while Debrezion, Pels and Rietveld (2007) confirmed through meta-analysis that railway stations can influence both residential and commercial property values. These studies are highly relevant to London because transport access is one of the most visible and practical determinants of where people choose to live.

At the same time, the literature suggests that transport effects are not always uniform. Banister and Thurstain-Goodwin (2011) argued that transport investment can generate wider non-transport benefits, while Yang et al. (2020) showed that the relationship between transit accessibility and property prices can vary spatially. Wittowsky et al. (2020) further found that accessibility, housing characteristics and neighbouring conditions all influenced residential prices in their case study. This is useful for the present study because it suggests that distance to the nearest station should not be treated as the only measure of accessibility. Broader locational context, including centrality and neighbourhood conditions, may also matter.

Neighbourhood context is particularly important in a city such as London, where socio-economic conditions vary strongly across areas. Although deprivation is not always included directly in older hedonic models, the wider literature on locational externalities and neighbourhood effects supports its inclusion. If areas differ in deprivation, services, environmental quality and local demand, then those differences are likely to be reflected in price.

This provides a clear justification for including IMD decile in the modelling framework. It also supports the idea that a house price model should incorporate both accessibility and neighbourhood conditions, rather than relying on property-level variables alone.

2.5 Machine learning approaches to house price prediction

More recent research has increasingly used machine learning methods to predict house prices because they can capture non-linear patterns and interactions more effectively than standard regression. This is particularly relevant for housing, where the effect of one variable may depend on another and where the relationship between price and location is rarely simple. Tree-based methods have become especially popular because they can handle mixed variable types and do not require strong linearity assumptions.

Rico-Juan and Taltavull de La Paz (2021) compared machine learning methods with spatial hedonic tools and found strong support for machine learning in the housing market of Alicante, Spain. Their study is especially useful because it combined predictive performance with explainability and showed that Random Forest could be both accurate and interpretable. Kim et al. (2022) also found that machine learning methods outperformed spatial interpolation methods in their comparative study of house prices. These studies support the use of non-linear models in the present dissertation. However, many such studies are based on different cities, institutional settings and data structures, which means that their conclusions cannot automatically be transferred to London. This creates a need for more city-specific evidence using London-focused open data.

The foundations of the models used in this dissertation are also well established in the wider machine learning literature. Random Forest was introduced by Breiman (2001), who described it as an ensemble of tree predictors whose performance depends on the strength of individual trees and the correlation between them. Gradient boosting was formalised by Friedman (2001), who showed how weak learners could be combined sequentially to improve performance. These methods are now widely used in predictive analytics and are suitable for housing data because they can handle complex interactions without needing strong parametric assumptions. Their use in this dissertation is therefore methodologically justified both by the housing literature and by the broader machine learning literature.

A baseline linear model is also useful in comparative modelling. Elastic Net, proposed by Zou and Hastie (2005), combines L1 and L2 regularisation and is often used when predictors may be correlated. Although it is less flexible than tree-based methods, it provides a useful benchmark because it shows what can be achieved using a more conventional regression-style approach with regularisation.

For this dissertation, the comparison between ElasticNet, Random Forest and Histogram Gradient Boosting provides a balanced modelling framework, combining baseline interpretability with more flexible non-linear methods.

2.6 Explainability in predictive housing models

As predictive models become more powerful, explainability becomes more important. In applied housing research, it is not enough to say that one model has lower error than another. Researchers also need to understand why a model performs well and which variables contribute most strongly to its predictions. This matters for academic interpretation, practical trust and policy relevance.

Lundberg and Lee (2017) introduced SHAP as a unified framework for interpreting model predictions. Their work is important because it provides a way to measure feature contributions in complex models and has become one of the most influential approaches in explainable machine learning. In the housing field, this issue has also been addressed directly by Teoh et al. (2023), who developed an explainable housing price prediction framework and emphasised the importance of identifying the determinants that influence housing values. These studies support the use of feature importance analysis in the present dissertation, since one of the project's aims is not only to compare predictive performance but also to understand which features matter most in the best-performing model.

This is particularly relevant because the present study includes engineered spatial and neighbourhood features such as distance to centre, distance to station and deprivation. If these features improve prediction, it is important to show whether they also emerge as influential predictors. Explainability therefore helps connect the technical results back to the broader literature on location, accessibility and neighbourhood context.

2.7 Neural networks and advanced approaches

Neural networks have also been used in house price research. Xu and Zhang (2021) showed that neural network models could perform effectively in a house price forecasting setting. Their work suggests that more complex non-linear models may be useful in some situations, especially when patterns in the data are difficult to capture using simpler methods. However, this does not mean that neural networks are always the best choice for structured tabular data. In many practical settings, tree-based models remain highly competitive and are often easier to interpret. For a dissertation, this matters because methodological clarity and explainability are important alongside predictive performance. This supports the decision in the present study to focus mainly on a linear baseline and tree-based ensemble models, while treating neural networks as a relevant but non-essential extension.

2.8 Summary of key themes in the literature

Table 1 Summary of key themes in the literature

Theme	Main idea from literature	Relevance to this dissertation
Hedonic pricing	House prices reflect a bundle of property and locational attributes	Provides the conceptual foundation for feature-based modelling
Spatial dependence	Housing values are influenced by geographic context and nearby areas	Justifies spatial variables such as coordinates and centrality
Transport accessibility	Distance to stations and transport links can affect property values	Supports inclusion of distance to the nearest station
Neighbourhood context	Deprivation and local area conditions influence housing value	Supports inclusion of IMD decile
Machine learning	Non-linear models often outperform simple linear models in housing prediction	Justifies comparing ElasticNet, Random Forest and Histogram Gradient Boosting
Explainability	Model interpretation is important alongside predictive performance	Supports use of feature importance and visual analysis

Table 1 shows that the literature converges around a common idea: house prices are shaped not only by the property itself, but also by spatial, accessibility and neighbourhood conditions. It also shows that modern predictive studies increasingly favour flexible machine learning methods while still emphasising the importance of interpretability. These themes directly inform the design of the present dissertation.

2.9 Research gap and relevance to the present study

The literature clearly shows that house prices are influenced by structural features, spatial dependence, accessibility and neighbourhood conditions, and that machine learning can improve prediction when compared with simpler models. It also shows that explainability is increasingly important in applied housing research. However, there is still room for studies that combine open transaction data with spatial and socio-economic features in a realistic forecasting framework.

Some house price studies use random train-test splits, which may produce optimistic results when the real aim is future prediction. Others focus strongly on model performance but give less attention to interpretability or to the practical value of spatially engineered features. In addition, many studies are based outside the UK, which limits their direct relevance to London. This dissertation addresses that gap by building a London-specific house price dataset from open sources, using a time-based split, comparing multiple machine learning regression models, and analysing the contribution of accessibility, centrality and deprivation to predictive performance. In this way, the study aims to combine theoretical relevance, practical modelling and interpretability within one framework.

3. Methodology

3.1 Introduction

This chapter explains the methods used in this study to predict house prices in London using machine learning and open data. The methodology was designed to answer the research questions in a structured and practical way. The study followed a quantitative approach and treated house price prediction as a supervised regression problem. This means that the models were trained to predict a continuous target value, namely residential sale price, using a set of property, spatial and neighbourhood-related variables.

The overall process included data collection, data cleaning, data integration, feature engineering, model development, model evaluation and interpretation. The main aim of the methodology was not only to compare the predictive performance of different models, but also to investigate whether spatial and socio-economic factors improve prediction and to understand which variables are most influential in the final model. For this reason, the methodology combined official transaction data with postcode geography, neighbourhood deprivation and transport accessibility information.

3.2 Research design

This study used a quantitative secondary data design. All data used in the analysis came from publicly available sources rather than primary data collection. This was appropriate because the dissertation aimed to model real house sale transactions using existing open datasets rather than collect perceptions, opinions or survey responses.

The project was also designed as a comparative modelling study. Instead of relying on only one algorithm, three regression models were tested and compared on the same dataset:

- ElasticNet
- Random Forest
- Histogram Gradient Boosting

This approach was chosen because it allowed comparison between a simpler linear model and more flexible non-linear models. It also made it possible to answer the first research question, which asked which machine learning model performs best for predicting London house prices.

The methodology also had a spatial dimension. Since location is a major determinant of housing value, the project did not rely only on basic property transaction variables. It also included spatial and neighbourhood-based features such as deprivation, distance to the nearest station and distance to Central London. This made the study more suitable for the London housing market and better aligned with the dissertation title.

3.3 Overall workflow of the study

The practical workflow of the study followed a staged pipeline. First, the HM Land Registry Price Paid Data for 2022, 2023 and 2024 were combined and reduced to the variables needed for modelling. Second, the transaction data were merged with the ONS Postcode Directory to obtain spatial coordinates and LSOA codes. Third, the merged dataset was filtered to London using a geographic bounding box and then cleaned further by removing implausible prices. Fourth, the model-ready London dataset was split into training and test sets using a time-based design. Fifth, the training and test files were enriched in stages by adding IMD decile, distance to the nearest station and distance to Central London. Finally, multiple regression models were trained, evaluated and interpreted.

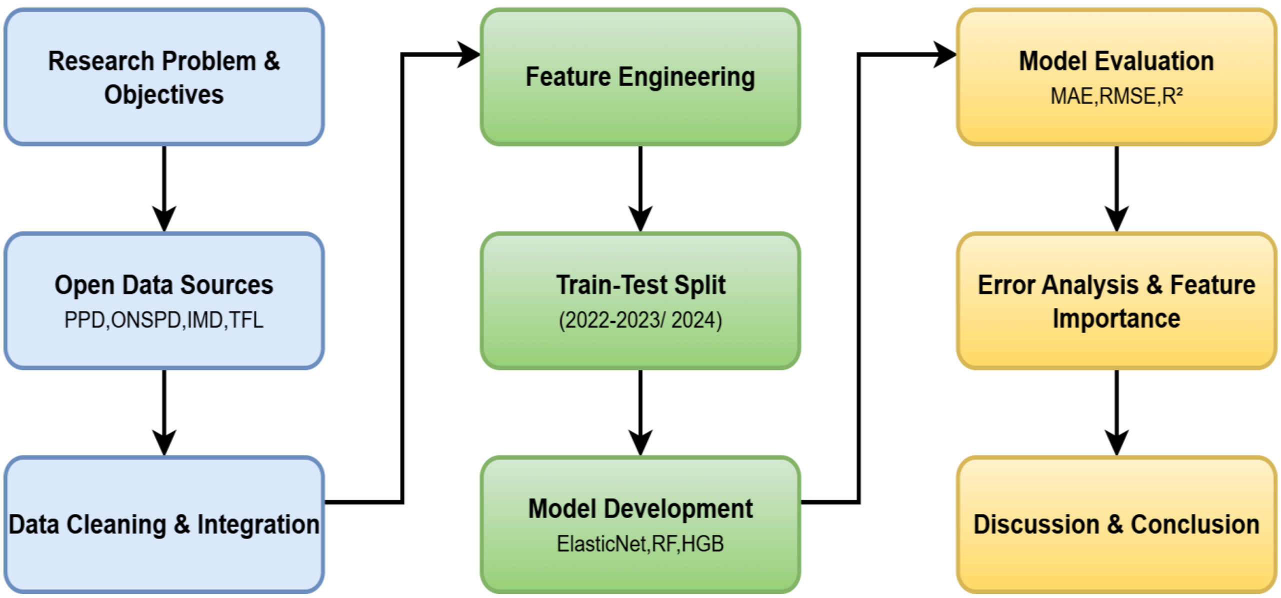

Figure 1 Workflow diagram of the study (created by author using diagrams.net)

Figure 1 summarises the study from data collection and cleaning through to feature engineering, modelling, evaluation and interpretation.

3.4 Data sources

Four main data sources were used in this study.

3.4.1 HM Land Registry Price Paid Data

The main dataset used in the dissertation was the HM Land Registry Price Paid Data (HM Land Registry, 2014). This dataset contains official records of residential property sales in England and Wales. For this project, the yearly files for 2022, 2023 and 2024 were used. These files provided the core transaction information, including:

- sale price
- date of transfer
- postcode
- property type
- new build flag
- tenure type

This dataset was chosen because it is an official and reliable source of real residential transaction data, and because it is widely used in UK property research.

3.4.2 ONS Postcode Directory (ONSPD)

The ONS Postcode Directory (Office for National Statistics, 2016) was used to link each transaction postcode to geographic information. The Land Registry data includes postcode, but it does not directly provide coordinates or neighbourhood identifiers. ONSPD was therefore used to add:

- latitude
- longitude
- Lower-layer Super Output Area (LSOA) code

This step was essential because it allowed the dataset to be spatially enriched and linked to neighbourhood-level deprivation data.

3.4.3 English Indices of Deprivation (IMD) 2019

The English Indices of Deprivation 2019 (Ministry of Housing, Communities and Local Government, 2019) were used to add a socio-economic context variable. The specific variable used in this project was IMD decile, which represents the relative deprivation level of an area. A lower decile indicates a more deprived area, while a higher decile indicates a less deprived area.

This variable was added because house prices are influenced not only by property characteristics but also by neighbourhood conditions. Including IMD allowed the model to capture an important part of local context.

3.4.4 Transport for London stop data

The final external source was Transport for London stop and station data (Transport for London, n.d.). This was used to create an accessibility-related feature by calculating the distance from each property to the nearest station. Since transport accessibility is highly important in London, this was considered a useful variable for house price prediction.

In addition to this, a second location feature, distance to Central London, was created using property coordinates and a central London reference point.

3.5 Data preparation and cleaning

The raw datasets could not be used directly for modelling, so several cleaning and preparation steps were carried out.

3.5.1 Combining the transaction files

The Price Paid Data files for 2022, 2023 and 2024 were combined into one dataset. Since the raw PPD files contain many columns, only the variables needed for the analysis were retained at this stage. These were:

- price
- date_of_transfer
- postcode
- property_type
- new_build_flag
- tenure_type

Columns such as address text and other administrative fields were not kept because they were not essential for the modelling approach used in this dissertation.

3.5.2 Data cleaning and standarisation

Several basic cleaning steps were applied:

- records with missing price, date_of_transfer or postcode were removed;
- postcodes were standardised by trimming spaces and converting to upper case;
- date_of_transfer was converted into datetime format;
- records with invalid prices, such as values less than or equal to zero, were removed.

These steps were necessary to make sure the data could be merged correctly and analysed reliably.

3.5.3 Merging with postcode geography

The cleaned transaction dataset was merged with ONSPD using postcode as the matching key. This added latitude, longitude and LSOA code to each transaction. This stage created a geographically enriched transaction dataset that could then be filtered to London and linked to deprivation information.

3.5.4 Filtering to London

Because the original PPD dataset covers England and Wales, it was necessary to isolate London transactions. This was done using a geographic bounding box based on latitude and longitude. Properties were kept only if they fell within the selected London coordinate range. This method provided a practical and transparent way to isolate London transactions from the wider national dataset.

3.5.5 Outlier treatment

The exploratory analysis showed that the raw price variable contained extreme values, including very low and very high prices. To reduce the influence of unrealistic or highly unusual transactions, a price filter was applied and only properties within the range £10,000 to £10,000,000 were retained for modelling. This step was important because extreme values can distort regression models and evaluation metrics, especially RMSE.

3.5.6 Dataset construction summary

After staged cleaning, merging and filtering, the final modelling dataset contained 388,881 observations. A time-based split produced 262,997 observations for the 2022–2023 training set and 125,884 observations for the 2024 test set. After IMD was merged, missing imd_decile values accounted for 5.513% of the training data and 5.838% of the test data. The nearest-station feature was created using 2,835 TfL stop and station records. The main construction stages and outputs are summarised in Table 2.

Table 2 Dataset Construction steps and outputs

Stage	Output / result	Notes
Reduced PPD transaction file	2,847,476 rows, 6 columns	Combined 2022–2024 Land Registry transaction data with core variables only
Reduced ONSPD file	2,706,777 rows, 4 columns	Retained postcode, LSOA, latitude and longitude
PPD merged with ONSPD	2,847,166 rows	Added spatial identifiers and coordinates
Initial London-only dataset	610,152 rows	Transactions retained after London filtering
London bounding-box dataset	391,763 rows	Refined London subset for modelling
Final model-ready dataset after outlier filter	388,881 rows	Cleaned dataset used for time-based split
Training set (2022–2023)	262,997 rows	Used for model training
Test set (2024)	125,884 rows	Used for final evaluation
Missing IMD in training set	5.513%	Handled using median imputation
Missing IMD in test set	5.838%	Handled using median imputation
TfL stop and station records	2,835	Used to create dist_station_km
Final enriched modelling files	train_2022_2023_imd_tfl_center.csv and test_2024_imd_tfl_center.csv	Final files used in modelling and evaluation

Table 2 summarises the main stages of dataset construction and feature enrichment. The modelling dataset was built progressively rather than in a single merge step. This helped maintain transparency, allowed each stage to be checked separately, and reduced the risk of hidden

processing errors. After the final outlier filter, the cleaned modelling dataset contained 388,881 observations, which were then split into 262,997 training cases for 2022–2023 and 125,884 test cases for 2024.

3.6 Feature engineering

Feature engineering was one of the most important parts of the methodology. The aim was to create variables that better represent the spatial and socio-economic structure of the London housing market.

3.6.1 Log-transformed target variable

The target variable used for modelling was not the raw sale price directly, but the log-transformed price (log_price). This transformation was applied because the distribution of raw house prices was highly right-skewed, with a small number of very expensive properties stretching the upper tail.

Using log_price had several advantages:

- it reduced skewness in the target variable;
- it reduced the influence of extreme values;
- it made the modelling process more stable;
- it is a common approach in house price modelling.

The models were therefore trained on log_price, but predictions were converted back into pounds for final evaluation and interpretation.

3.6.2 Time variables

From the transaction date, two time variables were created:

- year
- month

The year variable was mainly used for the train-test split, while month was included as a possible seasonal predictor.

3.6.3 Deprivation

Using the LSOA code, each property was matched to an IMD decile. This created the feature imd_decile, which was used as a neighbourhood-level socio-economic indicator. This was the first major enrichment stage after the initial time-based train-test split.

3.6.4 Distance to nearest station

After the training and test files had been enriched with IMD decile, a transport accessibility feature was added using TfL stop and station data. Property latitude and longitude were compared with TfL stop coordinates, and the nearest station was identified using a BallTree nearest-neighbour approach with haversine distance. The resulting variable was saved as dist_station_km, representing the distance in kilometres from each property to its nearest station.

3.6.5 Distance to Central London

After the station-distance feature was created, a second location-based feature was added to represent centrality. Using the property coordinates, the haversine distance from each property to a central London reference point was calculated. Charing Cross was used as the central reference, and the resulting feature was stored as dist_center_km. This feature was included because centrality is strongly associated with housing value in large urban markets and later proved to be one of the most important predictors in the final model.

3.7 Handling missing values

Most variables in the final dataset had little or no missing data. However, after merging IMD, around 5–6% of observations had missing imd_decile. Rather than dropping these records, missing numeric values were handled during the modelling stage using median imputation.

This approach was chosen because:

- it allowed the study to retain a large sample size;
- it is a simple and standard method for handling missing numeric values;
- it reduces the impact of missingness without introducing unnecessary complexity.

Median imputation was applied inside the preprocessing pipeline so that the same method was used consistently across models.

3.8 Train-test split

A time-based split was used rather than a random split. This was an important methodological decision because the aim was to test whether models trained on earlier transactions could predict later house prices.

The split was:

- training data: 2022 and 2023

- test data: 2024

This approach is more realistic than random splitting because it reflects a future prediction setting. It also reduces the risk of overly optimistic results that can happen when very similar transactions from the same period appear in both training and test data.

3.9 Models used

Three regression models were selected for comparison.

3.9.1 ElasticNet

ElasticNet was used as the baseline model. It is a regularised linear regression model that combines L1 and L2 penalties. It was included because it provides a simple and interpretable benchmark and is useful for showing whether more advanced non-linear methods offer clear improvement.

3.9.2 Random Forest

Random Forest was used as the main tree-based ensemble model. It consists of many decision trees whose predictions are averaged. This model was chosen because it is well suited to structured tabular data and can capture non-linear relationships and interactions between variables without requiring strong assumptions.

3.9.3 Histogram Gradient Boosting

Histogram Gradient Boosting was used as another non-linear model. This method builds trees sequentially to reduce prediction error and is often effective for structured datasets. It was included because boosting methods frequently perform strongly in machine learning tasks and provided a useful comparison with Random Forest.

3.10 Preprocessing pipeline

To ensure consistency across the models, a preprocessing pipeline was used.

The final modelling dataset contained both numeric and categorical variables. These are summarised in Table 3. Numeric variables were median-imputed where necessary. The categorical variables property_type, new_build_flag and tenure_type were transformed using one-hot encoding so that they could be represented in numerical form for the machine learning models. One-hot encoding converts each category into a separate binary indicator variable. This preprocessing was carried out within a scikit-learn pipeline to ensure that the same steps were applied consistently across all models.

Table 3 Predictor variables and target variable used for modelling

Variable	Type	Role	Description
price	Numeric	Original outcome	Raw residential sale price in pounds
log_price	Numeric	**Target variable**	Log-transformed sale price used for model training
lat	Numeric	Predictor	Property latitude
lon	Numeric	Predictor	Property longitude
month	Numeric	Predictor	Month of transaction
imd_decile	Numeric	Predictor	Neighbourhood deprivation decile
dist_station_km	Numeric	Predictor	Distance to nearest station in kilometres
dist_center_km	Numeric	Predictor	Distance to Central London in kilometres
property_type	Categorical	Predictor	Residential property type
new_build_flag	Categorical	Predictor	Indicates whether the property is a new build
tenure_type	Categorical	Predictor	Indicates tenure category, such as freehold or leasehold
year	Numeric	Split variable	Used for the time-based train-test split rather than as a final predictor

Table 3 summarises the final variables used in the modelling framework. The target variable for training was log_price, while price was retained for interpretation and reporting in pounds. The variable year was used for the time-based split rather than as a final predictor in the model.

3.11 Hyperparameter tuning

After the baseline model comparison, a light hyperparameter tuning stage was carried out to test whether model performance could be improved further. Because of the size of the dataset, a practical tuning strategy was used rather than a large exhaustive search. Baseline versions of ElasticNet, Random Forest and Histogram Gradient Boosting were first evaluated on the 2024 test set. A smaller tuning search was then applied to each model, and the tuned versions were re-evaluated on the same test set.

The purpose of tuning was not extensive optimisation, but to assess whether modest gains could be achieved within the scope of the dissertation. The tuning results showed that only limited improvements were obtained overall. The tuned Random Forest model remained the strongest final model, suggesting that the main gains in the dissertation came from data preparation and feature engineering rather than hyperparameter optimisation alone.

3.12 Evaluation metrics

Several evaluation metrics were used to assess model performance.

3.12.1 Mean Absolute Error (MAE)

$$MAE = (1/n) \times \Sigma |y_i - \hat{y}_i|$$

where y_i is the true price, $\hat{y}_i$ is the predicted price, and n is the number of observations

MAE measures the average absolute difference between predicted and actual price in pounds. It is easy to interpret and gives a clear idea of typical error size.

3.12.2 Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{(1/n) \times \Sigma(y_i - \hat{y}_i)^2}$$

RMSE also measures error in pounds, but gives more weight to larger errors. This is useful because it highlights whether the model makes occasional very large mistakes.

3.12.3 R^2

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$$

where $\bar{y}$ is the mean of the observed prices.

R^2 measures how much of the variation in price is explained by the model. It provides a general indication of model fit.

3.12.4 Median Absolute Error

$$MedianAE = median(|y_i - \hat{y}_i|)$$

Median absolute error was included because it gives a more robust measure of typical error and is less sensitive to very large outliers than MAE.

3.12.5 Within 10%

$$\textit{Within 10\%} = (1/n) \times \Sigma\, I(|y_i - \hat{y}_i| \leq 0.10 y_i) \times 100$$

where I is an indicator function equal to 1 if the condition is true and 0 otherwise.

This metric measures the percentage of predictions that fall within 10% of the true price. It was included because it is intuitive and easy to understand in practical terms.

These metrics were chosen because no single measure is sufficient on its own. Together they provide a more complete view of predictive performance.

3.13 Visualisation and interpretation

In addition to the numerical metrics, several visualisations were used to understand model behaviour. These included:

- raw price and log-price distributions;
- predicted versus actual values;
- residual distribution;
- error by price band;
- error by property type;
- correlation heatmap;
- permutation feature importance.

These plots were included because they make the results easier to interpret and help explain where the model performs well and where it struggles.

3.14 Software and tools

The analysis was carried out in Python using Google Colab. Google Drive was used for data storage. The main libraries used were:

- pandas for data handling and merging;
- numpy for numerical operations;

- scikit-learn for preprocessing, modelling, tuning and evaluation;
- matplotlib for visualisation.

This software environment was chosen because it allowed efficient data processing, model comparison and reproducible experimentation.

3.15 Summary

This methodology combined official residential transaction data with spatial and socio-economic variables to create a London house price prediction framework. The study used data cleaning, staged file construction, feature engineering, a realistic time-based split, multiple regression models and a light tuning stage in order to compare performance in a fair and structured way. Overall, the methodology was designed not only to identify the best-performing model, but also to show clearly how the dataset was built and how spatial and neighbourhood factors contributed to house price prediction in London.

4. Results

4.1 Introduction

This chapter presents the main findings of the study. The purpose of the analysis was to compare different machine learning models for predicting residential house prices in London using open data, and to assess whether spatial and neighbourhood-related features improved predictive performance. The chapter first presents the exploratory analysis of the final dataset, followed by the comparison of the three regression models used in the study. It then examines the effect of feature engineering and a light hyperparameter tuning stage before analysing the behaviour of the best-performing model in more detail using visualisations and feature importance.

4.2 Exploratory Data Analysis

Before comparing the machine learning models, it was necessary to examine the main characteristics of the final dataset. Exploratory data analysis was used to understand the target variable, inspect relationships between the numeric features, and support key methodological decisions such as the use of log-transformed price.

4.2.1 Final modelling dataset

The final modelling dataset was constructed from the HM Land Registry Price Paid Data for 2022, 2023 and 2024, then enriched with postcode geography, deprivation and transport-accessibility features. After London filtering and outlier treatment, the final analytical dataset contained 388,881 observations. A time-based split was then applied, producing 262,997 training observations for 2022–2023 and 125,884 test observations for 2024. The final predictors included geographic coordinates, month, IMD decile, distance to the nearest station, distance to Central London, property type, new build status and tenure type. The target variable used for model training was log_price, while final performance was reported in pounds using the original price scale.

4.2.2 Distribution of raw house prices

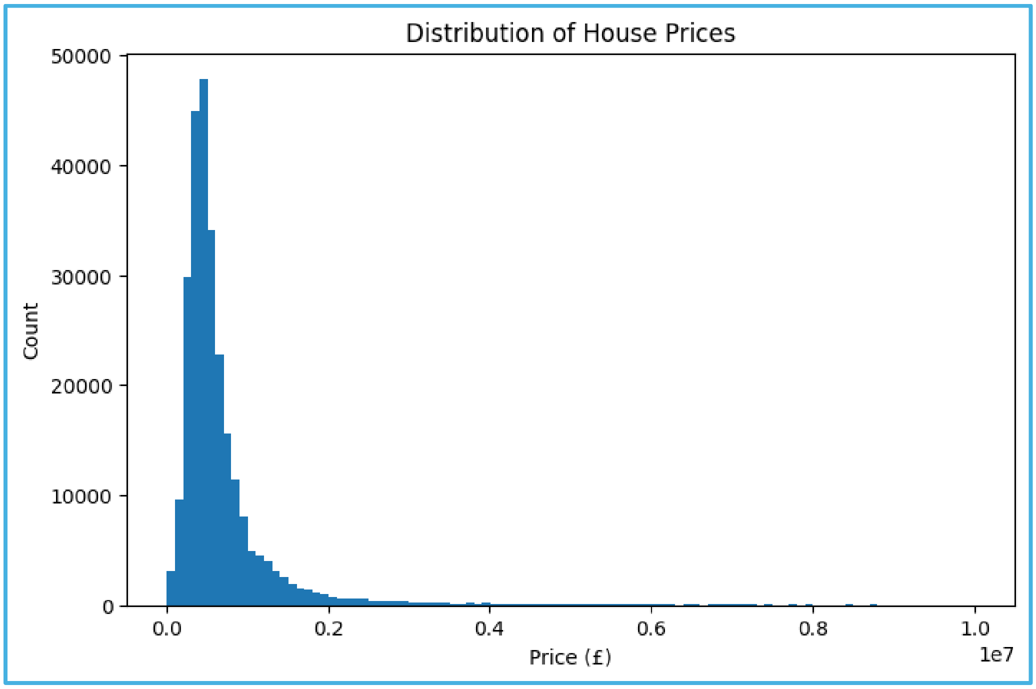

Figure 2 Distribution of House Prices

Figure 2 shows the distribution of raw house prices in the training data. The distribution is strongly right skewed, with most transactions concentrated in the lower and middle price ranges and a long upper tail created by a smaller number of expensive properties.

This is a common pattern in house price data, especially in London, where a large number of ordinary residential sales exist alongside a much smaller number of premium and luxury transactions. The figure shows that the raw price variable is far from normally distributed. This is important because a highly skewed target can make modelling more difficult, especially when a small number of large values dominate the learning process.

4.2.3 Distribution of log-transformed prices

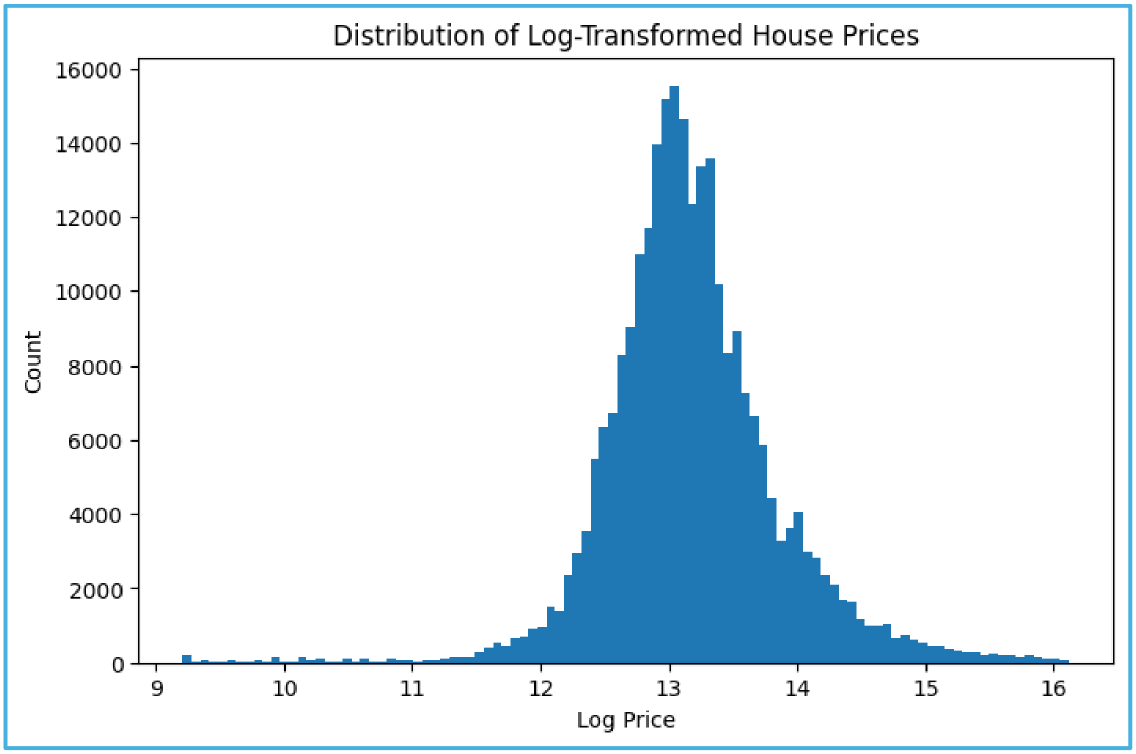

Figure 3 Distribution of Log-Transformed House Prices

Figure 3 shows the distribution of log_price. Compared with the raw price distribution, the log-transformed target is much more balanced and closer to a bell-shaped form. The upper tail is compressed, and the values are more evenly spread across the central range.

This figure provides visual support for the decision to use log_price rather than raw price in the modelling stage. The purpose of the transformation was to reduce skewness and reduce the influence of extreme high-value properties. The transformed distribution suggests that this choice was appropriate.

Together, Figures 2 and 3 justify one of the key methodological decisions in the dissertation. The raw target variable was highly skewed, whereas the log-transformed version was much more suitable for regression modelling.

4.2.4 Correlation analysis of numeric variables

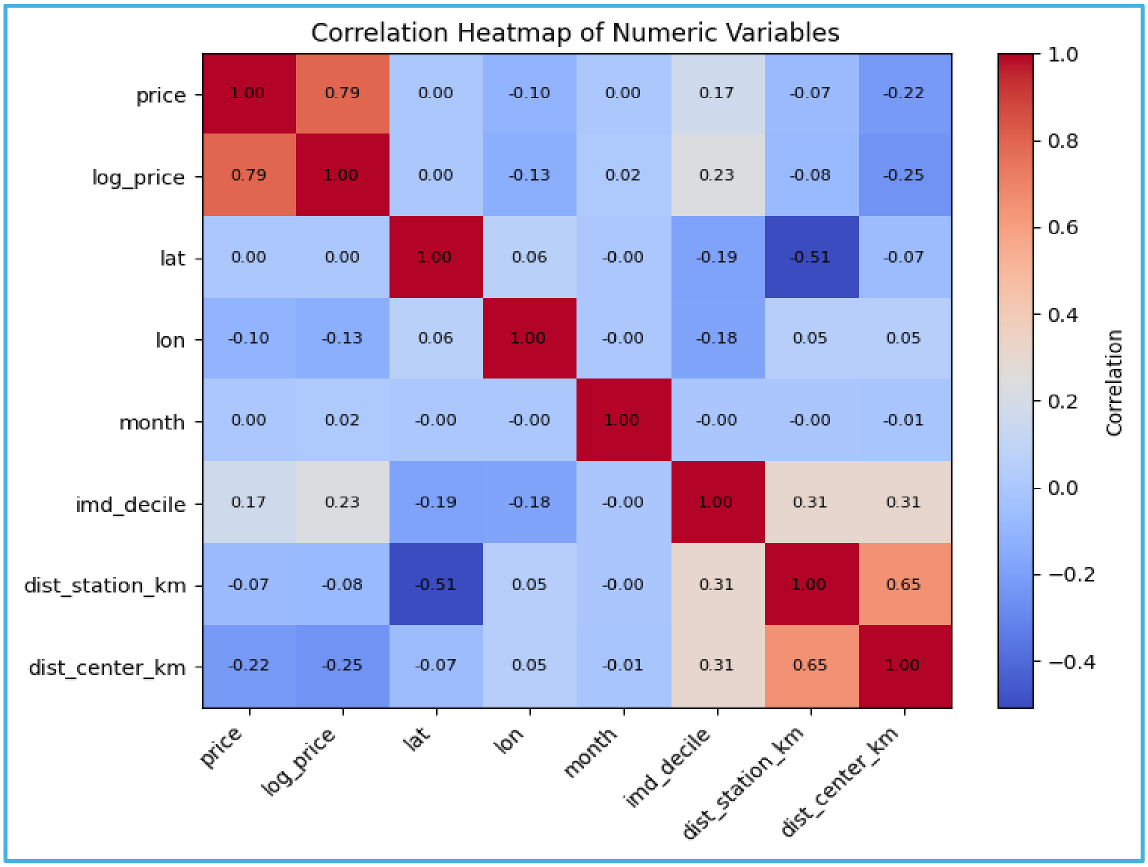

Figure 4 Correlation Heatmap of Numeric Variables

A correlation heatmap was created for the main numeric variables in the training set. This was used to explore broad linear relationships between the variables before the modelling stage.

The heatmap shows that price and log_price are strongly positively correlated, which is expected because one is a transformation of the other. More importantly, dist_center_km has the strongest negative relationship with both price and log_price among the engineered features. This suggests that properties located further from Central London generally tend to have lower sale prices.

The heatmap also shows a positive relationship between imd_decile and both price variables. Since higher IMD deciles represent less deprived areas, this suggests that properties in less deprived neighbourhoods tend to have higher prices. This is a meaningful result and provides early support for the inclusion of deprivation as a feature in the modelling process.

By contrast, month shows almost no linear relationship with price. This suggests that seasonal effects are weak in the selected dataset, at least in simple correlation terms. dist_station_km shows a weak negative relationship with price, which means that properties further from stations tend to be slightly cheaper on average, although the linear relationship is not especially strong.

The heatmap also shows that dist_station_km and dist_center_km are moderately correlated. This is not surprising, since properties further from the centre may also tend to be further from major stations. Overall, the correlation analysis suggests that centrality and deprivation are likely to be more important than simple seasonality, which is consistent with the broader spatial structure of the London housing market.

4.3 Model comparison

The next stage of the analysis compared the three selected machine learning models:

- ElasticNet
- Random Forest
- Histogram Gradient Boosting

All three models were trained on the same training set and evaluated on the same 2024 test set.

4.3.1 Final model comparison

Table 4 Final model comparison on the 2024 test set

Model	MAE (£)	RMSE (£)	R^2	MedianAE (£)	Within 10% (%)
ElasticNet	253,576.69	622,839.81	0.2099	121,029.00	22.10
Random Forest	191,716.64	502,547.67	0.4856	80,728.21	33.09
HistGradientBoosting	217,084.17	547,225.83	0.3901	96,632.58	27.93

Table 4 shows that Random Forest was the best-performing model overall. It achieved the lowest MAE, the lowest RMSE, the highest R^2, the lowest median absolute error, and the highest percentage of predictions within 10% of the true sale price.

This means that Random Forest was the most effective model for the London housing prediction problem in this study.

4.3.2 ElasticNet results

ElasticNet produced the weakest overall performance of the three models. Its MAE was approximately £253,577, and its R^2 was 0.2099. This indicates that the model explained only a relatively small proportion of the variation in test-set prices.

This result suggests that a regularised linear model was not flexible enough to capture the complexity of the relationships in the London housing market. Given the strong spatial variation and likely non-linear interactions between features, this is not surprising. However, ElasticNet still played an important role in the study because it provided a baseline against which the stronger non-linear models could be compared.

4.3.3 Random Forest results

Random Forest achieved the strongest overall results. Its MAE was approximately £191,717, and its RMSE was around £502,548. The R^2 value of 0.4856 indicates that the model explained nearly half of the variation in the test-set house prices.

The median absolute error was £80,728, which is much lower than the MAE. This suggests that the typical prediction error was considerably smaller than the average error, and that some larger mistakes were pulling the MAE upward. The Within 10% value of 33.09% means that roughly one in three predictions fell within 10% of the true price.

These results indicate that Random Forest was the most balanced and effective model in the study. It was able to capture more of the variation in price than the other models and was consistently stronger across all evaluation measures.

4.3.4 Histogram Gradient Boosting results

Histogram Gradient Boosting also performed well, but not as well as Random Forest. Its MAE was approximately £217,084, its RMSE was £547,226, and its R^2 was 0.3901. These results are clearly better than ElasticNet, but weaker than Random Forest.

This suggests that Histogram Gradient Boosting was also able to capture non-linear relationships in the data, but did not fit the structure of this dataset as effectively as Random Forest. Even so, it remained a strong model and showed that more flexible machine learning methods outperform simple linear approaches for this problem.

4.3.5 Overall interpretation of the comparison

The model comparison suggests that non-linear tree-based models are better suited to London house price prediction than a linear baseline model. This implies that the relationships between the predictors and the target are not simple or fully linear. Instead, they likely involve interactions, thresholds and location-specific effects that are better captured by ensemble tree methods.

This is one of the key findings of the dissertation because it directly answers the first research question. Among the models tested, Random Forest performed best.

To understand whether the final results were driven by hyperparameter optimisation or by the addition of engineered variables , the next two sections examine the effect of tunning and feature engineering separately.

4.3.6 Effect of hyperparameter tuning

After the baseline model comparison, a light hyperparameter tuning stage was carried out to test whether model performance could be improved further

Table 5 Baseline and tuned model performance on the 2024 test set

Model	Version	MAE (£)	RMSE (£)	R^2	MedianAE (£)	Within 10% (%)
ElasticNet	Baseline	253,574.29	623,153.31	0.2091	120,989.18	22.13
ElasticNet	Tuned	253,576.69	622,839.81	0.2099	121,029.00	22.10
Random Forest	Baseline	192,281.83	504,103.52	0.4824	80,602.15	33.42
Random Forest	Tuned	191,716.64	502,547.67	0.4856	80,728.21	33.09
HistGradientBoosting	Baseline	217,327.58	546,293.66	0.3922	97,203.42	27.77
HistGradientBoosting	Tuned	217,084.17	547,225.83	0.3901	96,632.58	27.93

The tuning results show that hyperparameter optimisation produced only modest changes overall. The tuned Random Forest remained the strongest model and achieved slight improvements in MAE, RMSE and R^2. However, the improvements were small, and tuning did not materially change the overall interpretation of the results.

This suggests that the main gains in the study came from feature engineering and dataset construction rather than from extensive optimisation alone.

4.4 Effect of feature engineering

A major part of the study involved adding new spatial and neighbourhood-related variables to the dataset. These included:

- imd_decile
- dist_station_km
- dist_center_km

The aim of this stage was to test whether these features improved predictive performance and made the final model more meaningful in spatial terms.

4.4.1 Performance before and after adding engineered features

Table 6 Effect of feature engineering on Random Forest performance

Model version (Random Forest)	MAE (£)	RMSE (£)	R^2	MedianAE (£)	Within 10% (%)
Basic feature set	194,503.68	506,390.80	—	—	—
+ IMD and station distance	193,355.76	507,553.21	0.4753	81,035.96	33.11
+ IMD, station distance and centre distance	192,281.83	504,103.52	0.4824	80,602.15	33.42

Note: R^2, MedianAE and Within 10% were not recorded for the basic feature-set stage, so only MAE and RMSE are reported for that comparison. This table reports baseline Random Forest performance before hyperparameter tuning.

The results show that feature engineering produced incremental improvements. Adding IMD and distance to station slightly improved MAE, although RMSE became marginally worse. Once dist_center_km was added, both MAE and RMSE improved again, and the final Random Forest model achieved its best baseline performance before tuning.

The numerical gains were not extremely large, but they were still useful. More importantly, the engineered variables strengthened the conceptual quality of the model by making it more spatially informed. The fact that these variables later appeared as important predictors also confirms that they added meaningful information rather than unnecessary complexity.

4.4.2 What this suggests

This result suggests that the final model benefited from spatial and neighbourhood-based features, even if the gains in headline metrics were not dramatic. The strongest addition was clearly dist_center_km, which later became the most important variable in the model. This shows that centrality matters in London housing prices and that feature engineering contributed not only to performance but also to interpretability.

4.5 Behaviour of the best-performing model

Because Random Forest was the strongest model overall, the next part of the analysis focuses on its behaviour in more detail.

4.5.1 Predicted versus actual prices

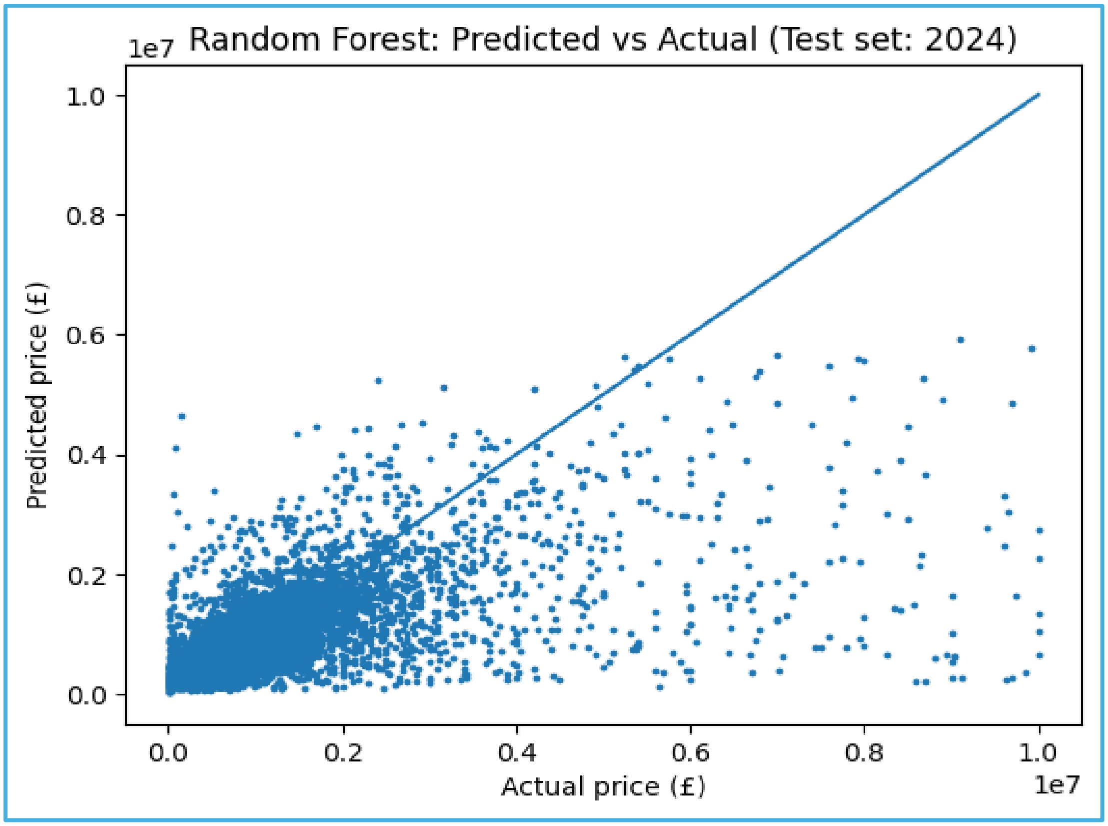

Figure 5 Random Forest: Predicted vs Actual (Test set: 2024)

Figure 5 compares the actual sale prices in the test set with the prices predicted by the Random Forest model. A clear positive relationship can be seen, which shows that the model captures the general price structure of the market reasonably well.

Many observations lie close to the diagonal reference line, especially in the lower and middle parts of the market. However, as the actual price increases, the spread of the points becomes larger and many observations fall below the line. This means that the model tends to underpredict higher-value properties.

This is an important finding because it shows that the model is not equally accurate across the full range of the housing market. It performs better for ordinary transactions than for luxury properties. This is likely to reflect both the smaller number of very expensive observations and the absence of some features that are important for premium homes, such as floor area, number of bedrooms, condition and detailed micro-location characteristics.

4.5.2 Residual distribution

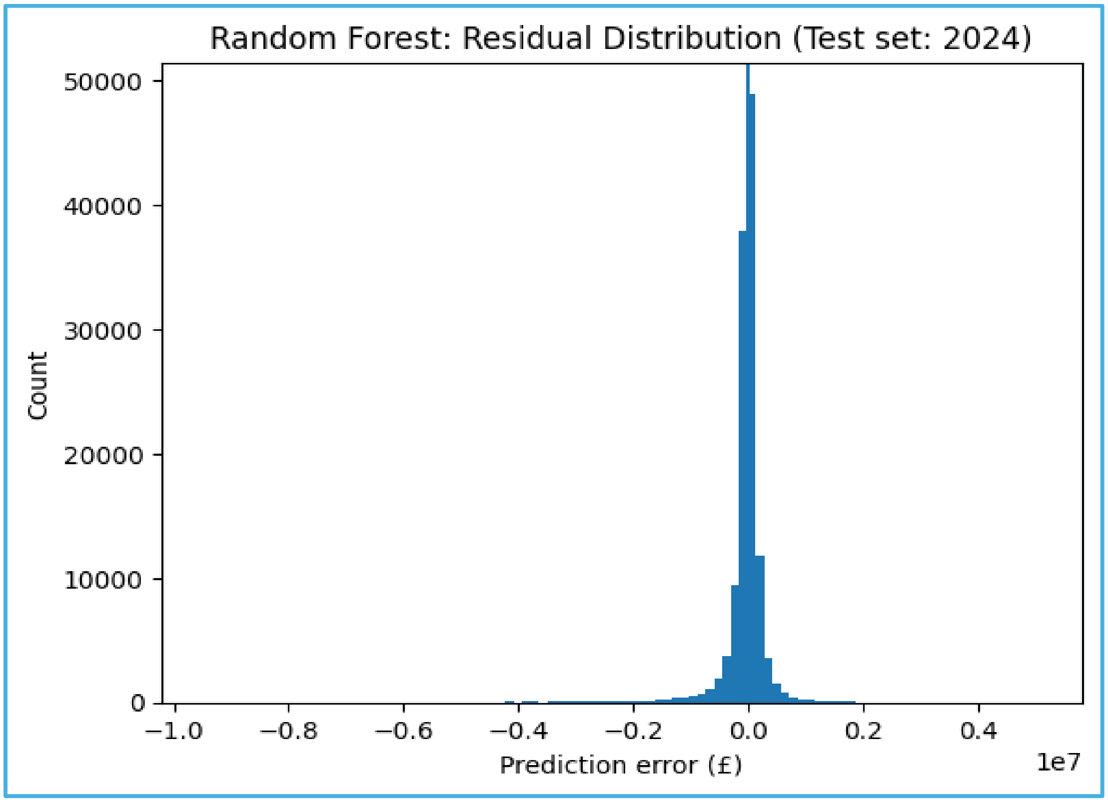

Figure 6 Random Forest: Residual Distribution

The residual distribution provides another way of examining model error. In this study, error was calculated as predicted price minus actual price, meaning that:

- values near zero indicate accurate prediction;
- negative values indicate underprediction;
- positive values indicate overprediction.

The residual plot shows that many observations are concentrated close to zero, which suggests that a large proportion of predictions are reasonably accurate. However, the distribution is not perfectly symmetrical. It contains long tails, especially on the negative side, which indicates the presence of some large underpredictions.

This supports the pattern seen in the predicted-versus-actual plot. The Random Forest model does not fail across the whole market, but it does struggle more with certain high-value or unusual cases.

4.6 Error analysis

To understand model behaviour more clearly, additional error analysis was carried out.

4.6.1 Error by price band

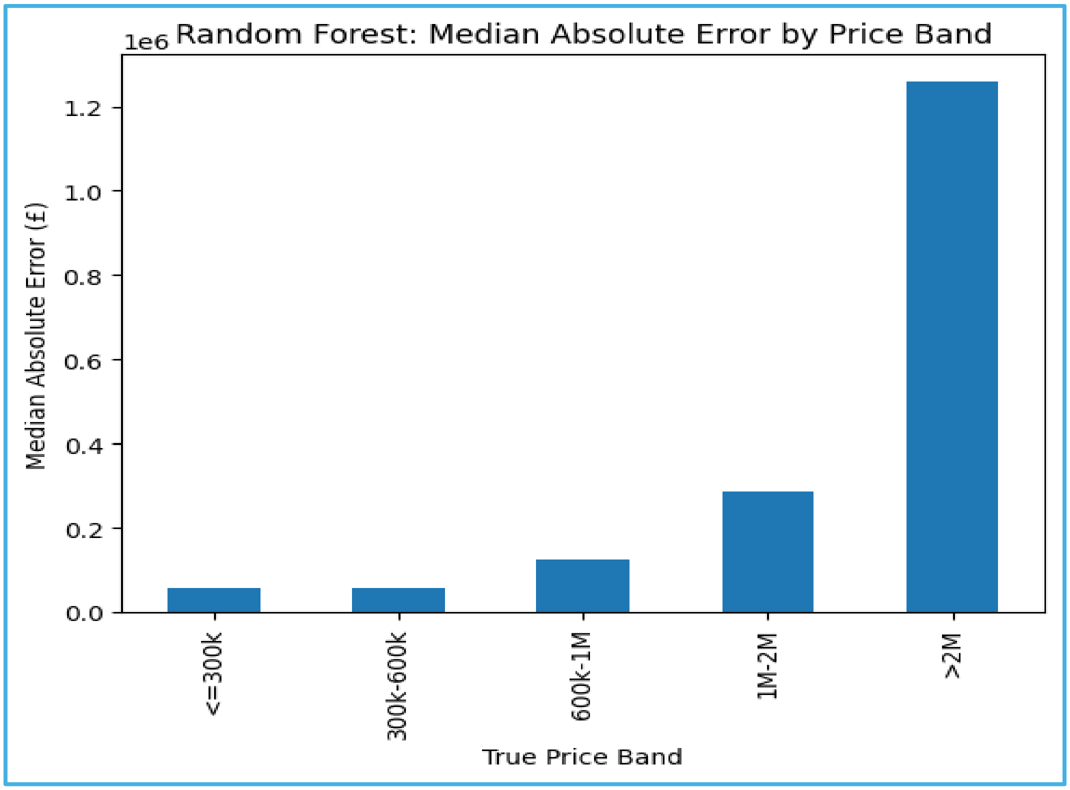

Figure 7 Random Forest: Median Absolute Error by Price Band

Figure 7 shows that median absolute error increases sharply as the price band increases. The model performs relatively well for lower- and middle-priced properties, but errors become much larger for more expensive homes. The highest price band, above £2 million, has by far the greatest median error.

This result strongly suggests that the model is best suited to the mainstream London housing market rather than the upper end. This is understandable because luxury properties are fewer, more diverse, and likely to be influenced by variables that are not included in the current dataset.

This figure is especially valuable because it shows where the model is strong and where it is weak. It adds detail that cannot be seen from the overall MAE alone.

4.6.2 Error by property type

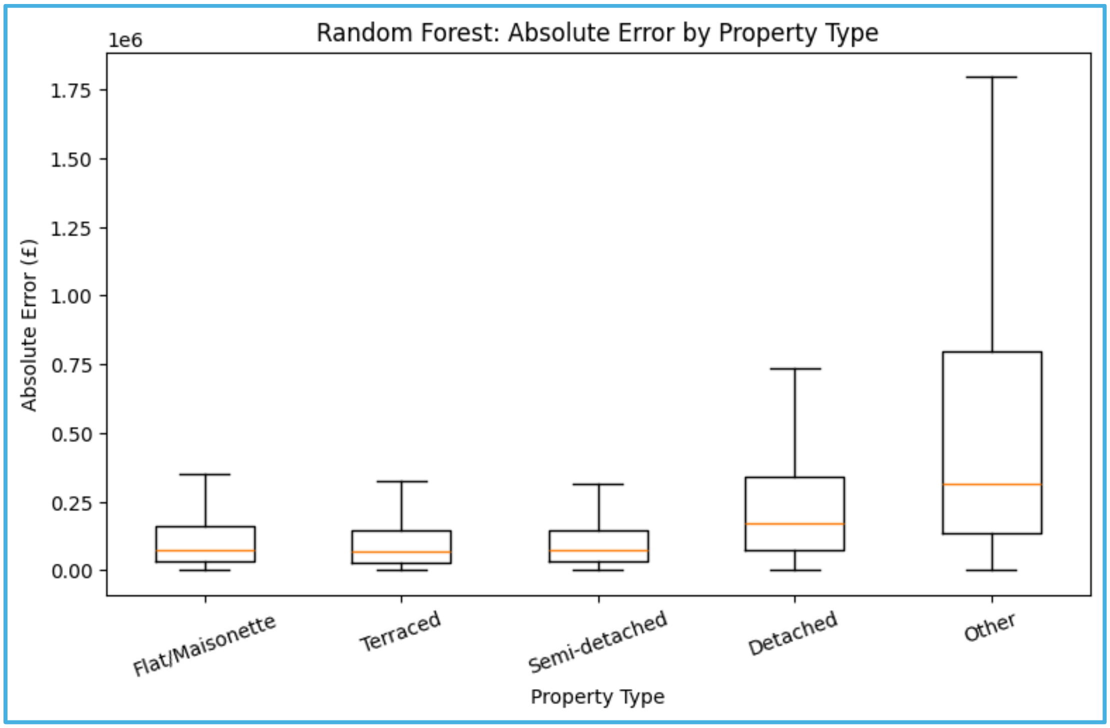

Figure 8 Random Forest: Absolute Error by Property Type

Median absolute error by property type was:

- Terraced: approximately £67,000
- Semi-detached: approximately £72,263
- Flat/Maisonette: approximately £75,719
- Detached: approximately £168,945
- Other: approximately £314,167

The plot shows that the model performs best for terraced houses, semi-detached houses and flats/maisonettes. Detached properties have much higher error, while the "other" category is by far the most difficult to predict.

This suggests that the model performs better for common and relatively regular property types than for more unusual or expensive categories. Detached and "other" properties are often more heterogeneous and may be affected by features not captured in the dataset.

This result fits closely with the price-band analysis and reinforces the conclusion that the model is more accurate for mainstream properties than for premium or less typical homes.

4.7 Feature importance

4.7.1 Permutation feature importance

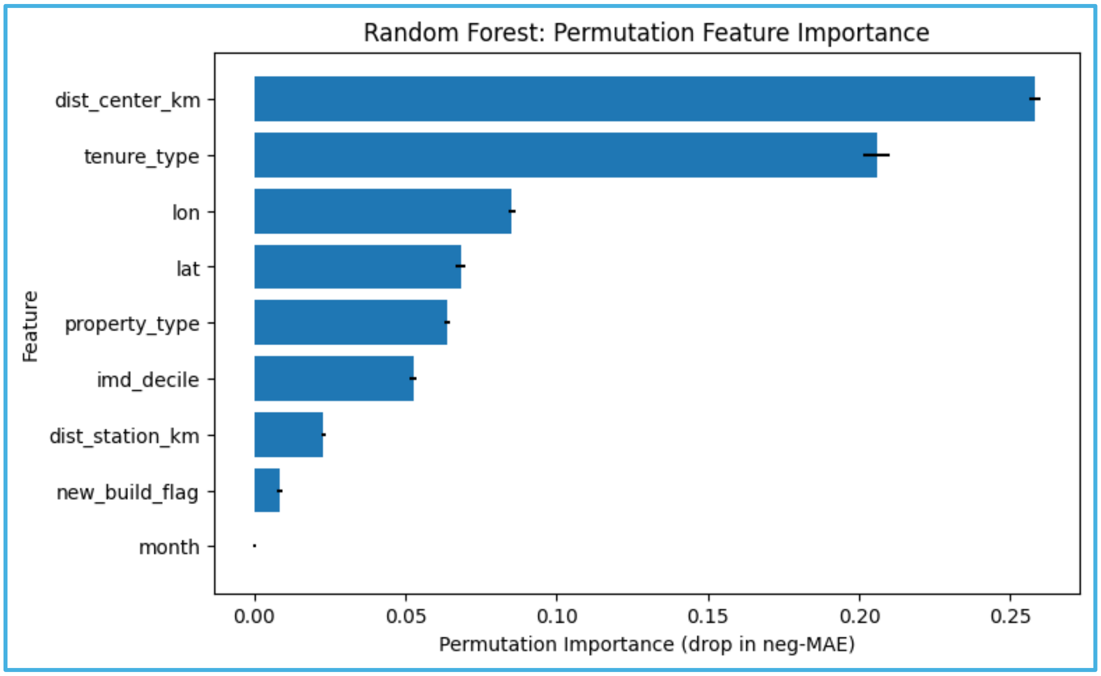

Figure 9 Random Forest: Permutation Feature Importance

The permutation feature importance results show the following ranking:

1. dist_center_km
2. tenure_type
3. lon
4. lat
5. property_type
6. imd_decile
7. dist_station_km

8. new_build_flag
9. month

The most important variable by a clear margin was distance to Central London. This is one of the most significant findings in the dissertation because it suggests that centrality is the strongest driver of price in the final model.

Tenure type was the second most important feature, which indicates that leasehold/freehold differences have a strong effect in the London housing market. Longitude and latitude were also highly important, showing that location still matters strongly even after adding engineered spatial features.

property_type and imd_decile also contributed meaningfully, while dist_station_km had a smaller but still positive effect. The low importance of month suggests that seasonal timing had little influence compared with spatial and structural variables.

4.7.2 Interpretation

The feature importance results are valuable because they show that the final model is not simply a black box. The model is mainly relying on:

- centrality,
- tenure,
- geographic location,
- property type,
- neighbourhood deprivation.

This supports the argument that housing values in London are shaped strongly by spatial position and neighbourhood context. It also confirms that the feature engineering carried out in this project was useful, particularly the creation of dist_center_km, which became the most important variable in the final model.

4.8 Summary of the results

The main findings of this chapter can be summarised as follows.

First, Random Forest was the strongest model among the three compared in the study. It achieved the best performance across all major regression metrics, and a light tuning stage improved it slightly further.

Second, the addition of spatial and neighbourhood-based features improved the quality of the modelling framework. In particular, distance to Central London proved to be a highly informative variable.

Third, the visual analysis showed that model performance was not uniform across all parts of the market. The Random Forest model performed much better for lower- and middle-priced homes than for expensive properties. Detached and unusual property types were also harder to predict.

Fourth, the feature importance analysis showed that centrality, tenure, location, property type and deprivation were among the most important drivers of prediction, while month had very little importance.

Overall, the results suggest that open data and machine learning can be used to predict London house prices reasonably well, but also that prediction remains more difficult for the most expensive and unusual parts of the market. These findings are discussed in the next chapter in relation to the literature, the research questions and the limitations of the study.

5. Discussion

5.1 Introduction

This chapter discusses the main findings of the study in relation to the research questions and the literature reviewed earlier. The purpose is not simply to repeat the results, but to explain what they mean and what they suggest about house price prediction in London using open data and machine learning.

Overall, the findings show that machine learning can be used effectively for London house price prediction when the modelling framework includes spatial and neighbourhood-based features. At the same time, the results also show that predictive performance is weaker for expensive and unusual properties, which reflects the limits of the available data.

5.2 Which model performed best?

The first research question asked which machine learning model performed best for predicting London house prices using the selected open datasets. The results showed that Random Forest was the strongest model. After light hyperparameter tuning, it achieved a MAE of £191,716.64, RMSE of £502,547.67, R^2 of 0.4856, Median Absolute Error of £80,728.21, and 33.09% of predictions within 10% of the true price.

This suggests that the London house price problem is better captured by a non-linear tree-based ensemble model than by a regularised linear model. ElasticNet performed worst, which indicates that the relationships in the data are more complex than a linear structure can represent. Histogram Gradient Boosting also performed reasonably well, but did not outperform Random Forest.

A light tuning stage was carried out for all three models. However, the gains were modest. The tuned Random Forest improved slightly in MAE, RMSE and R^2, but the overall model ranking did not change. This suggests that the main improvements in the dissertation came from dataset construction and feature engineering, rather than from hyperparameter optimisation alone.

5.3 Did spatial and socio-economic features improve prediction?

The second research question asked whether spatial and socio-economic features improved predictive performance. The results suggest that they did, although the improvement was moderate rather than dramatic.

The addition of IMD decile, distance to the nearest station, and especially distance to Central London improved the overall modelling framework. The strongest addition was clearly dist_center_km, which later became the most important variable in the Random Forest model. This indicates that centrality remains one of the strongest influences on housing value in London.

The positive role of imd_decile also suggests that neighbourhood socio-economic context matters. Properties in less deprived areas tended to be associated with higher values, which supports the wider literature on neighbourhood effects. Similarly, dist_station_km contributed positively, although less strongly than dist_center_km, indicating that transport accessibility matters but is not the dominant location effect in the final model.

These findings show that feature engineering was important not only for model performance, but also for interpretability. Even where the numerical gains were limited, the added variables made the modelling framework more meaningful in housing terms.

5.4 Which variables were most influential?

The third research question asked which variables appeared to have the strongest influence on the predictions of the best-performing model. The permutation feature importance results showed that the most influential variables were:

- distance to Central London
- tenure type
- longitude
- latitude
- property type
- IMD decile
- distance to nearest station
- new build flag
- month

The strongest feature by a clear margin was distance to Central London, which suggests that centrality is one of the key drivers of value in the London housing market. This is a meaningful finding because it links directly to both urban geography and practical housing logic. Properties closer to central areas often provide better access to jobs, services, infrastructure and established high-value locations.

The second most important variable was tenure type, which is also important in the London market, where freehold and leasehold structures can strongly affect value. Latitude and longitude remained influential even after adding engineered spatial variables, suggesting that raw location still contains important local information beyond broader centrality.

Overall, the feature importance results show that the final model was shaped mainly by centrality, tenure, location, property type and neighbourhood deprivation. This supports the argument that housing value in London depends strongly on spatial and neighbourhood context rather than on property type alone.

5.5 Why did the model struggle with expensive and unusual properties?

The visual and error analysis showed that the model performed less accurately for expensive homes and unusual property types. This was visible in the predicted-versus-actual plot, the residual distribution, the price-band error plot and the property-type error plot.

There are several likely reasons for this. First, expensive properties are often more heterogeneous than ordinary homes. Second, they are less common in the data, which gives the model fewer examples from which to learn. Third, the dataset did not include several variables that are likely to matter strongly for premium housing, such as floor area, number of rooms, condition, parking, garden space and detailed micro-location prestige.

This means that the model relied mainly on broader location and neighbourhood signals. That is sufficient for modelling the mainstream market reasonably well, but less effective for highly distinctive or luxury properties. This should not be treated as a failure of the dissertation, but as an important finding about the limits of open-data housing prediction.

5.6 Strengths and limitations

The study has several strengths. It uses official open datasets, combines multiple spatial and neighbourhood data sources, applies a realistic time-based train-test split, and evaluates models using several metrics rather than one alone. It also goes beyond simple prediction by interpreting feature importance and model behaviour.

At the same time, the study has clear limitations. The main limitation is the lack of detailed property-level variables such as floor area, number of bedrooms and condition. A second limitation is the use of a London bounding box rather than a formal administrative boundary. A third limitation is that only three models were compared, and tuning was deliberately kept light rather than extensive.

5.7 Final reflection

Taken together, the findings suggest that machine learning can be used productively for London house price prediction when the modelling framework includes meaningful spatial and neighbourhood features. The results also show that the largest gains came from feature engineering and dataset construction rather than tuning alone.

From a practical point of view, the final model is most useful for broad residential market prediction rather than detailed valuation of luxury properties. From an academic point of view, the dissertation demonstrates that combining open data, spatial feature engineering, model comparison and explainability can produce a strong and meaningful machine learning study.

6. Conclusion

6.1 Overview

This dissertation examined whether machine learning models, supported by open spatial and socio-economic data, could be used to predict residential house prices in London. The study also compared multiple regression models and assessed the contribution of neighbourhood deprivation, transport accessibility and centrality to predictive performance.

Overall, the dissertation showed that machine learning can be used effectively for London house price prediction, especially when the modelling framework includes spatially meaningful features. At the same time, the study also showed that the performance of open-data models is weaker for expensive and unusual properties.

6.2 Main findings

The first research question asked which model performed best. The results showed that the tuned Random Forest was the strongest model, achieving a MAE of £191,716.64, RMSE of £502,547.67, R^2 of 0.4856, Median Absolute Error of £80,728.21, and 33.09% of predictions within 10% of the true price on the 2024 test set.

The second research question asked whether spatial and socio-economic features improved prediction. The answer was yes. The addition of IMD decile, distance to the nearest station, and especially distance to Central London improved the modelling framework and made the final model more meaningful in spatial terms.

The third research question asked which features were most influential in the best-performing model. The results showed that the final model was driven mainly by distance to Central London, tenure type, geographic coordinates, property type and neighbourhood deprivation. This suggests that London house prices are shaped strongly by centrality, ownership structure, local spatial position and socio-economic context.

6.3 Overall conclusion

Taken together, the findings suggest that London house price prediction benefits from combining machine learning with engineered spatial and neighbourhood variables. The study demonstrated that open data can be used to create a practical and interpretable modelling pipeline, and that non-linear tree-based models are better suited to this problem than a simple linear baseline.

However, the dissertation also showed that the model was much more effective for lower- and middle-priced homes than for the luxury segment. This reflects both the complexity of expensive properties and the absence of richer property-level variables such as floor area, room count and condition.

6.4 Limitations and future work

The main limitation of the study was the lack of detailed property-level information. Future work could improve the model by adding variables such as floor area, number of rooms, energy performance and property condition, and by using more precise geographic boundaries. Further research could also compare additional algorithms and explore more advanced explainability methods.

6.5 Final statement

In conclusion, this dissertation has shown that open data spatial machine learning can provide a useful and academically valuable framework for predicting house prices in London. Among the models tested, Random Forest performed best, while centrality, tenure and neighbourhood context emerged as key influences on housing value. Although the model has clear limitations, especially for high-value properties, the study demonstrates that combining machine learning with spatial thinking is a strong and relevant approach to understanding the London housing market.

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Appendix

1) Data_Set_Files

Note on code files: The original .ipynb file was created in Google Colab across multiple sessions during the dissertation process, so it contains repeated imports, repeated Google Drive mounting commands, and some intermediate working cells. For clarity, the separate Notepad/text code file has been cleaned by removing repeated library imports, repeated drive-mount commands, and one incomplete exploratory tuning block. Both files have been uploaded. These changes were made only to improve readability and do not affect the final methodology, analysis, or results.

Dataset construction stages and output files

Stage	Output file	Description
1	ppd_2022_2024_min.csv	Combined and reduced transaction data from HM Land Registry
2	onspd_min.csv	Reduced postcode geography dataset
3	ppd_onspd_2022_2024.csv	Transaction data merged with postcode geography
4	ppd_london_bbox_2022_2024.csv	London-only subset using geographic bounding box
5	london_model_ready_2022_2024.csv	Model-ready London dataset before time split
6	train_2022_2023.csv / test_2024.csv	Time-based split into training and test data
7	train_2022_2023_imd.csv / test_2024_imd.csv	IMD decile added
8	train_2022_2023_imd_tfl.csv / test_2024_imd_tfl.csv	Distance to nearest station added
9	train_2022_2023_imd_tfl_center.csv / test_2024_imd_tfl_center.csv	Distance to Central London added

Figure A1: Colab output for reduced PPD(price paid data) transaction dataset creation

```
Saved: /content/drive/MyDrive/DS7010_HousePrices/data/processed/ppd_2022_2024_min.csv
Rows, Cols: (2847476, 6)
```

	price	date_of_transfer	postcode	property_type	new_build_flag	tenure_type
0	280000	2022-06-06	B78 3XA	D	N	F
1	312500	2022-06-20	ST7 4JZ	D	N	F
2	150000	2022-06-30	ST14 7QG	S	N	F
3	515000	2022-05-25	B78 3UG	D	N	F
4	262500	2022-07-15	ST20 0HY	D	N	F

Figure A2: Colab output for reduced ONSPD postcode geography dataset creation

```
Total columns in ONSPD: 53
Using columns: pcds lat long lsoa01cd
Saved: /content/drive/MyDrive/DS7010_HousePrices/data/processed/onspd_min.csv
Rows, Cols: (2706777, 4)
```

	postcode	lsoa	lat	lon
0	AB1 0AA	S01000011	57.101459	-2.242858
1	AB1 0AB	S01000011	57.102539	-2.246315
2	AB1 0AD	S01000011	57.100541	-2.248349
3	AB1 0AE	S01000333	57.084429	-2.255714
4	AB1 0AF	S01000007	57.096641	-2.258109

Figure A3: Colab output for merged PPD-ONSPD dataset creation

```
print("Saved:", out_path)
print("PPD rows:", len(ppd))
print("ONSPD rows:", len(onspd))
print("Merged rows:", len(merged))
merged.head()
```

```
Saved: /content/drive/MyDrive/DS7010_HousePrices/data/processed/ppd_onspd_2022_2024.csv
PPD rows: 2847476
ONSPD rows: 2706777
Merged rows: 2847166
```

	price	date_of_transfer	postcode	property_type	new_build_flag	tenure_type	lsoa	lat	lon
0	280000	2022-06-06	B78 3XA	D	N	F	E01029839	52.617397	-1.700116
1	312500	2022-06-20	ST7 4JZ	D	N	F	E01029582	53.107444	-2.207053
2	150000	2022-06-30	ST14 7QG	S	N	F	E01029430	52.904086	-1.889535
3	515000	2022-05-25	B78 3UG	D	N	F	E01029488	52.599563	-1.719358
4	262500	2022-07-15	ST20 0HY	D	N	F	E01029709	52.788000	-2.257516

Figure A5: Colab output for final cleaned dataset and time-based train-test split

```
# Save splits
train_path = os.path.join(PROC, "train_2022_2023.csv")
test_path  = os.path.join(PROC, "test_2024.csv")
train.to_csv(train_path, index=False)
test.to_csv(test_path, index=False)

print("Saved:", train_path)
print("Saved:", test_path)
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
After outlier filter rows: 388881
Train rows (2022-2023): 262997
Test rows (2024): 125884
Train price range: 10000 to 10000000
Test price range: 10000 to 10000000
Saved: /content/drive/MyDrive/DS7010_HousePrices/data/processed/train_2022_2023.csv
Saved: /content/drive/MyDrive/DS7010_HousePrices/data/processed/test_2024.csv
```